



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

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TITLE OF THE PROJECT		
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Abstract:

Smart Item Recovery Portal AI Driven Smart Matching Management System is an important requirement in educational institutions, organizations, and corporate campuses where users frequently lose or discover personal belongings. Traditional lost and found systems generally depend on manual reporting, record maintenance, and physical searching, which can be time-consuming, inefficient, and prone to errors. The lack of an intelligent mechanism for automatically identifying relationships between lost and found item descriptions can further delay the recovery process. Therefore, there is a need for a secure, scalable, and intelligent lost and found management system that can automate item management and improve the chances of successful recovery.

To address these challenges, this project proposes a Secure AI-Powered Lost & Found Management System using Microservices Architecture. The system is developed using Spring Boot and follows Service-Oriented Architecture (SOA) principles, where individual business functionalities are implemented as independent and loosely coupled microservices. The proposed architecture consists of User Service, Item Service, AI Matching Service, Notification Service, API Gateway, and Eureka Service Discovery Server. These services communicate through RESTful APIs and can be independently developed, deployed, maintained, and scaled according to system requirements.

The User Service manages user registration, authentication, and authorization. Security is provided using Spring Security and JSON Web Tokens (JWT) to ensure that only authenticated and authorized users can access protected resources. The Item Service manages lost and found item information, including item descriptions, categories, locations, dates, and other relevant details. Spring Cloud Eureka is used for service discovery, enabling microservices to register themselves and dynamically discover other available services. Spring Cloud Gateway provides a centralized entry point for client requests and performs request routing to the appropriate microservice.

The major feature of the proposed system is the AI Matching Service, which assists users in identifying potentially matching lost and found items. When a user reports a lost item, the service analyzes the textual description using Natural Language Processing (NLP) techniques. TF-IDF (Term Frequency-Inverse Document Frequency) is used to represent the important terms in item descriptions, while Cosine Similarity is applied to measure the similarity between lost and found item descriptions. Based on the calculated similarity scores, the system identifies potential matches and provides relevant matching information to users, thereby reducing manual searching and improving the efficiency of item recovery.

The system also incorporates a Notification Service to inform users about relevant item matches and important updates. The microservices architecture improves modularity, maintainability, scalability, and fault tolerance because each service can be developed and deployed independently. The project follows Agile development practices, including sprint planning, user stories, task management, testing, and continuous integration. Unit and integration testing can be performed using JUnit 5 and Spring Boot Testing Frameworks, while GitHub Actions can be used to support CI/CD automation.

Collectively, the proposed system combines AI-based matching, secure authentication, service discovery, API gateway routing, and microservices architecture to provide an efficient and reliable solution for lost and found management. The system aims to reduce the time required to identify lost items, minimize manual effort, protect user and item information, and improve the overall recovery process. The proposed solution can be effectively deployed in educational institutions, corporate campuses, and other large organizations where secure and scalable lost and found management is required.

Project Architecture/Project Overview:

